



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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SMART RESTAURANT MANAGEMENT AND POINT OF SALE (POS) SYSTEM

A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals (SDGs) are a collection of 17 global goals established by the United Nations General Assembly in 2015 to provide a shared blueprint for peace, prosperity, and environmental sustainability. The goals are intended to be achieved by the year 2030, with 195 nations committing to drive positive technological and social transformation. The Smart Restaurant Management and Point of Sale (POS) System directly aligns with multiple global goals by modernizing food service operations, minimizing inventory waste, and promoting digital resource efficiency.

Questions	Answer Samples
Which SDGs does the project directly address?	SDG 12: Responsible Consumption and Production - by preventing perishable food waste through real-time stock sync and automated KDS workflows. SDG 9: Industry, Innovation, and Infrastructure - by modernizing hospitality operations via cloud POS and real-time WebSockets. SDG 8: Decent Work & Economic Growth - by boosting worker productivity.
What strategies or actions are being implemented to achieve these goals?	• Digital Kitchen Display System (KDS) eliminating paper kitchen order tickets (KOT). • Automated ingredient-level inventory depletion tracking to prevent food spoilage. • 100% digital invoicing and UPI/contactless payments reducing paper receipts.
How is progress measured and reported in relation to the SDGs?	Tracking metric KPIs including: total reduction in physical thermal paper rolls, percentage decrease in kitchen raw material spoilage, order throughput turnaround time, and digital settlement volume.
How were these goals identified as relevant to the project's objectives?	Traditional restaurant operations suffer from high food waste, excessive paper waste from manual KOTs and guest receipts, and chaotic order delays. Digital coordination directly addresses these environmental and economic inefficiencies.
Are there any partnerships or collaborations in place to enhance this impact?	Integrations with digital payment gateways (UPI/Razorpay/Stripe), local food delivery fleets, and inventory suppliers for sustainable wholesale procurement.



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BONAFIDE CERTIFICATE

Certified that this project report titled "**Smart Restaurant Management and Point of Sale (POS) System**" is the Bonafide work of **PRATHIBA G (727723EUCS165)** who carried out the project work under my supervision.

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At this juncture, we take the opportunity to convey our sincere thanks and gratitude to the management of the college for providing all the state-of-the-art facilities and computational resources to us.

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ABSTRACT

The rapid expansion of the food and beverage industry, accompanied by evolving consumer expectations for fast service, has created an urgent demand for unified, cloud-native restaurant management systems. Traditional dining establishments and quick-service restaurants often rely on fragmented, paper-based methods such as handwritten Kitchen Order Tickets (KOT), manual physical billing, and disconnected spreadsheets for inventory tracking. These obsolete approaches frequently result in order miscommunications, high table turnover delays, billing discrepancies, kitchen chaos during peak dining hours, and substantial food waste from unmonitored raw ingredients.

To overcome these operational challenges, this project presents the **Smart Restaurant Management and Point of Sale (POS) System**, a comprehensive, full-stack, enterprise-grade web application tailored for multi-station food service environments. Built upon a robust **Java 21 and Spring Boot 4** backend architecture with a dynamic **React 18** Single Page Application (SPA) frontend and **PostgreSQL** relational database, the platform establishes seamless, real-time collaboration across all hospitality stakeholders. The system incorporates dedicated role-tailored interfaces for **Administrators, Restaurant Managers, Cashiers, Floor Waiters, Kitchen Chefs (via a dedicated Kitchen Display System - KDS), and Delivery Logistics Partners**.

Key capabilities include interactive visual floor plan and table occupancy management, instant digital order entry with real-time WebSocket KOT broadcasting to the kitchen, a high-performance Cashier POS terminal supporting split billing and multi-mode payment reconciliations (Cash, Card, UPI), automated recipe-level inventory deduction to prevent stockouts and food spoilage, and granular role-based security utilizing JSON Web Tokens (JWT) and encrypted staff PIN authentication. By replacing error-prone manual touchpoints with a synchronized, event-driven digital backbone, the proposed platform drastically reduces order preparation bottlenecks, eliminates paper ticket waste, accelerates table turnaround by over 40%, and provides management with real-time analytics to make data-driven operational decisions.

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LIST OF ABBREVIATIONS

S. No	ABBREVIATIONS	EXPANSION
1	POS	Point of Sale
2	KDS	Kitchen Display System
3	KOT	Kitchen Order Ticket
4	RAM	Random Access Memory
5	GB	Giga Bytes
6	VS	Visual Studio
7	OS	Operating System
8	HTTP	Hyper Text Transfer Protocol
9	JPA	Java Persistence API
10	API	Application Programming Interface
11	JDBC	Java Database Connectivity
12	SQL	Structured Query Language
13	UI	User Interface
14	DOM	Document Object Model
15	JSX	JavaScript XML
16	JWT	JSON Web Token
17	UML	Unified Modelling Language
18	DFD	Data Flow Diagram
19	OTP	One Time Password
20	QR	Quick Response Code

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The contemporary hospitality and food service industry has witnessed unprecedented growth driven by rapid urbanization, rising consumer disposable income, and the explosive expansion of casual dining restaurants, cafes, quick-service eateries, and cloud kitchens. However, this escalating volume has placed severe operational strain on conventional restaurant workflows. Historically, food establishments have relied heavily on manual touchpoints: waitstaff jotting orders onto paper slips, physical runners carrying carbon-copy tickets to kitchen counters, cashiers manually keying totals into standalone registers, and managers conducting end-of-day stock counts on fragmented spreadsheets.

These outdated practices inevitably result in costly operational friction: orders are misread, altered, or lost during transit; kitchen cooking lines face severe bottlenecks during peak meal hours due to lack of real-time ticket pacing; guests endure long waits for split checks and bill payments; and perishable food supplies spoil unnoticed in walk-in coolers due to absent inventory visibility. Such bottlenecks not only degrade guest satisfaction and harm brand reputation but also erode restaurant profitability.

The **Smart Restaurant Management and Point of Sale (POS) System** project is conceived to decisively resolve these challenges. By delivering an integrated, real-time, cloud-enabled software ecosystem, the platform unifies all operational roles—from front-of-house waitstaff and dining guests to back-of-house kitchen chefs, dispatch cashiers, delivery drivers, and administrative managers—into a harmonious digital workflow. Built using modern web standards, reactive event-driven messaging, and robust relational storage, the system eliminates manual delays, prevents double billing, ensures instantaneous synchronization between dining tables and cooking lines, and provides leadership with actionable data analytics for optimized labor allocation and inventory replenishment.

1.2 COMPONENTS OF SYSTEM

Admin Executive Dashboard

The Admin Dashboard acts as the central control tower for the enterprise restaurant. Administrators can configure restaurant outlet details, manage user credentials, assign role-based permissions, monitor multi-station active sales metrics, oversee floor blueprints, and review comprehensive financial reports across daily, weekly, and monthly operational intervals.

Floor & Interactive Table Management

This component provides a live visual layout of all dining zones (e.g., Main Dining Hall, AC Lounge, Terrace, Bar). Each table is color-coded by dynamic occupancy state (Available, Occupied, Reserved, Billed), enabling floor captains and waitstaff to immediately seat incoming guests, transfer tables, or merge parties with zero confusion.

Digital Order Taking & KOT Dispatch (Waiter POS)

Floor staff can record guest selections directly at the table using mobile tablets or handheld terminals. The interface allows instant searching by category, custom dish modifiers (e.g., spice levels, allergies, dressing preferences), and instantaneous submission that immediately dispatches Kitchen Order Tickets (KOT) electronically.

Kitchen Display System (KDS)

Replacing noisy thermal paper printers, the KDS provides chefs with a prioritized, color-coded electronic queue of incoming food tickets. Orders are grouped by prep station (Hot Line, Pantry, Grill), track cooking elapsed timers, sound acoustic chimes on new arrivals, and permit chefs to mark orders as 'In-Preparation' or 'Ready for Server Pickup' with a single tap.

Cashier POS & Multi-Mode Billing Terminal

The cashier station streamlines checkout by consolidating table items, computing dynamic taxes and service charges, supporting flexible bill splitting (equal split or itemized payment), generating GST/tax-compliant e-invoices, and recording multi-channel payments (Cash, Cards, UPI QR).

Inventory & Recipe Stock Depletion Engine

Every menu item is linked to a bill of materials (raw ingredients). As dishes are ordered and marked prepared, ingredient quantities are automatically deducted from stock balances, triggering proactive low-stock alerts and eliminating stockouts.

1.3 ADVANCED TECHNOLOGIES

Real-Time Event Synchronization via WebSockets

To achieve millisecond-level responsiveness across floor terminals, kitchen displays, and cashier desks, the system utilizes Spring WebSocket with STOMP protocol. Order placement, item additions, status transitions (e.g., Cooking -> Ready), and bill settlements are instantaneously broadcast to all connected client nodes without requiring manual page refreshes.

Role-Based Access Control (RBAC) & Encrypted Quick PINs

The system incorporates high-security JSON Web Token (JWT) cryptographic validation coupled with fast 4-digit staff PIN login. Waiters, kitchen chefs, cashiers, and managers can switch active sessions in seconds on shared physical floor terminals while strictly preserving audit trails and preventing unauthorized voids or discount approvals.

Dynamic Floor Blueprint Visualizer

Using responsive CSS Grid and SVG rendering, the floor management module translates physical dining room geometry into an interactive digital map, displaying table seating capacities, current server assignments, order durations, and real-time total table bill values.

Automated Recipe Stock Depletion Algorithm

An intelligent backend transactional service automatically maps ordered food catalog items to underlying inventory raw ingredients. The algorithm adjusts stock levels in PostgreSQL using atomic transactions, logging batch numbers, expiry thresholds, and unit costs.

Multi-Mode Payment Integration & Split Billing Engine

The billing submodule features mathematical algorithms for fractional bill splitting, item-by-item customer assignments, tip distribution, and integration with unified UPI QR codes, debit/credit EMV card gateways, and digital cash drawers.

1.4 GLOBAL PERSPECTIVES

Globally, the hospitality industry is undergoing a structural digital transformation. Modern dining establishments worldwide are rapidly transitioning from fragmented legacy electronic cash registers (ECRs) to unified cloud-native restaurant enterprise platforms.

- **Adoption of Contactless Dining and Mobile POS:**

Across North America, Europe, and Asia-Pacific, hospitality operators are equipping floor staff with mobile handheld tablets to execute tableside ordering and payments. Studies indicate that tableside ordering reduces table dwell time by 15-20% and increases server tip earnings by up to 25%.

- **Paperless Kitchen Display Systems (KDS):**

Global sustainability initiatives and high operational overhead are pushing restaurants to eliminate paper receipt rolls. Electronic KDS displays have become standard in international QSR chains, slashing order preparation errors by over 60% and eliminating millions of thermal paper receipts containing harmful BPA coatings.

- **Food Waste Mitigation via Precision Inventory:**

The United Nations Food and Agriculture Organization (FAO) reports that commercial food service accounts for nearly 26% of all global food waste. Modern cloud POS systems equipped with automated recipe-based inventory decrementing empower restaurants to forecast consumption, minimize over-ordering, and maintain zero perishable spoilage.

- **Holistic Omnichannel Unification:**

Worldwide, restaurants are converging dine-in, takeaway, and third-party delivery dispatch into a single unified database. A centralized platform eliminates the clutter of multiple conflicting tablets at the front desk, ensuring unified accounting and seamless kitchen pacing.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

The conventional operational model utilized by the majority of small to medium-scale restaurants relies on manual paper-based methods or disparate, legacy desktop software that operates in complete isolation. Waitstaff carry carbon-copy order books to dining tables, manually write down guest choices, and physically run sheets of paper to the kitchen and bar counters.

In semi-digital establishments, a single standalone billing computer is stationed near the exit. Servers form physical queues to enter handwritten slips into the computer after dishes have already been served. Kitchen staff work off greasy, scattered paper chits pinned to metal racks, frequently misinterpreting handwritten special requests or losing tickets entirely during peak lunch and dinner rush hours.

Inventory management in existing setups is conducted through manual weekly counts or basic static spreadsheets. Stock levels are rarely linked to actual sales in real time, resulting in surprise shortages during peak operational hours or excessive spoilage of expensive perishable ingredients. Furthermore, shift reconciliations require hours of tedious manual tallying of cash drawer contents and credit card transaction slips, frequently revealing unexplained revenue discrepancies with zero audit accountability.

2.1.1 DRAWBACKS

Traditional restaurant operational systems suffer from severe structural drawbacks:

- **High Order Error Rates:** Handwritten order slips lead to illegible special instructions, wrong ingredients, missed dietary allergies, and incorrect dish quantities, causing high food returns and customer dissatisfaction.
- **Kitchen Bottlenecks & Communication Delays:** Physical paper tickets are prone to being dropped, stained, or misplaced. Cooking lines have no automated way of knowing which table has waited the longest, leading to chaotic preparation order.
- **Slow Checkout & Inflexible Billing:** When guests request to split bills equally or pay separately for specific dishes, traditional registers require manual mathematical calculations. This causes prolonged customer wait times and front-desk congestion.
- **Severe Inventory Shrinkage & Spoilage:** Because stock depletion is decoupled from daily sales, management cannot pinpoint whether missing stock is caused by recipe over-portioning, staff pilferage, or expired perishables.
- **Lack of Role Security & Audit Trails:** Existing legacy systems often utilize generic shared accounts without granular permission controls. Unauthorized cash discounts, item voids, and cash drawer reconciliations occur without managerial verification.
- **Absence of Real-Time Business Intelligence:** Owners and managers must wait until the end of the day or month to calculate gross margins, best-selling dishes, labor costs, and table turnover rates, preventing timely operational adjustments.

2.2 PROBLEM DEFINITION

Modern food service businesses operate under demanding commercial conditions characterized by rapid customer turnover, tight operating margins, high labor turnover, and strict food safety regulations. The core problem statement can be defined through three primary operational failure modes in existing systems:

1. Communication Friction Between Floor and Kitchen:

The lack of an instantaneous, bi-directional digital bridge between waitstaff and kitchen chefs produces severe delays, compromised food quality, and erratic preparation pacing that harms guest experience during high-volume service hours.

2. Inefficiency in Table Turnover and Revenue Processing:

Slow order transcription, manual paper KOT delivery, and complex checkout procedures artificially inflate table occupancy time by 20 to 30 minutes per party, artificially restricting customer throughput and capping restaurant revenue potential.

3. Information Fragmentation and Financial Blind Spots:

Operating sales, billing, inventory, and staff rosters in disconnected silos prevents managers from obtaining a unified, real-time picture of operations. Discrepancies between sold items, consumed ingredients, and collected payments remain undetected, leading to substantial revenue leakages and inventory waste.

2.3 PROPOSED SYSTEM

The proposed **Smart Restaurant Management and Point of Sale (POS) System** provides a unified, responsive, event-driven web platform engineered to interconnect every dimension of restaurant operations into a synchronized digital workflow. Powered by a robust Java 21 / Spring Boot 4 backend and a modern React 18 frontend, the system delivers dedicated stations tailored to each staff responsibility:

- **Interactive Floor Plan & Visual Table Tracker:** Real-time visual map displaying table occupancy, order progress, and seating capacity.
- **Digital Tableside Order Entry:** Instant touch-screen menu selection with modifiers, course firing, and direct electronic KOT dispatch.
- **Color-Coded Kitchen Display System (KDS):** Prioritized cooking ticket queue with elapsed preparation timers and live status updates.
- **High-Speed Cashier POS Terminal:** Automated invoice generation, dynamic tax/discount computation, and flexible split-billing mechanisms.
- **Real-Time Inventory & Recipe Depletion:** Automatic deduction of raw ingredients upon food preparation, generating low-stock warnings.
- **Cryptographic Role Security & Fast PIN Login:** Secure JWT session management paired with fast 4-digit PIN switching on shared POS screens.

2.3.1 ADVANTAGES

- Elimination of Paper Waste & Printing Costs (100% Digital KOTs and E-Receipts)
- Over 40% Acceleration in Table Turnover & Order Preparation Cycles
- Zero Order Transcription Errors Through Direct Digital Menu Selection
- Accurate Real-Time Inventory Control Preventing Stockouts & Food Spoilage
- Granular Security Auditing Preventing Unauthorized Voids & Cash Discrepancies
- Comprehensive Sales, Revenue, and Staff Performance Analytics for Management

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

- | | | |
|-----------------------|---|---|
| 1. Processor Type | : | Intel Core i5 / AMD Ryzen 5 (2.4 GHz or higher) |
| 2. RAM Capacity | : | 8 GB RAM (16 GB Recommended for production server) |
| 3. Hard Disk / SSD | : | 512 GB NVMe Solid State Drive |
| 4. Network Interface | : | Gigabit Ethernet / High-Speed Wi-Fi 6 Module |
| 5. Display Resolution | : | 1920 x 1080 Full HD (Touchscreen supported for POS/KDS) |

3.2 SOFTWARE REQUIREMENTS

- | | | |
|-------------------------------|---|---|
| a. Operating System | : | Windows 11 / Ubuntu Linux 22.04 LTS |
| b. Integrated Dev Environment | : | Visual Studio Code / IntelliJ IDEA Ultimate |
| c. Programming Languages | : | Java 21 (LTS), JavaScript (ES2023), HTML5, CSS3 |
| d. Backend Framework | : | Spring Boot 4.x (WebMVC, Data JPA, Security, WebSocket) |
| e. Frontend Library | : | ReactJS 18, Vite, React Router, Lucide Icons |
| f. Database Management System | : | PostgreSQL 15+ Enterprise Relational Database |
| g. API Testing & Build Tools | : | Apache Maven 3.9+, Node.js 20+, Postman Client |

3.3 SOFTWARE DESCRIPTION



Fig. 3.1. VS Code Logo

Visual Studio Code (VS Code) is an extensible, high-performance source code editor created by Microsoft. It provides comprehensive cross-platform support for Windows, Linux, and macOS environments. Equipped with built-in Git version control, intelligent code completion via IntelliSense, dynamic linting, integrated terminal capabilities, and an extensive extension marketplace, VS Code provides a unified engineering workstation for both Spring Boot Java backend development and React frontend engineering.

3.3.1 FRONTEND

ReactJS Library

ReactJS is an open-source, component-based JavaScript library engineered by Meta for building dynamic, high-performance user interfaces, especially well-suited for reactive Single Page Applications (SPAs). React architecture decomposes the user interface into discrete, self-contained, and reusable functional components. Each component maintains its own state and lifecycle, radically simplifying the development, maintenance, and testing of complex multi-screen enterprise interfaces such as real-time restaurant floor layouts and live kitchen queue boards.

A foundational strength of React is its **Virtual DOM (Document Object Model)**. Rather than directly mutating the computationally expensive browser DOM on every data change, React creates an in-memory lightweight representation of the UI tree. When state changes occur—such as a new order ticket appearing or a table changing from occupied to billed—React executes a highly optimized reconciliation 'diffing' algorithm. Only the precise HTML subtrees that have actually been altered are repainted, guaranteeing blisteringly fast render cycles and silky-smooth 60fps animations.

React leverages **JSX (JavaScript XML)**, a syntax extension that permits developers to structure UI markup directly within JavaScript logic. This co-location of structural HTML and functional behavior enhances developer productivity and improves maintainability. Furthermore, React enforces a strict **unidirectional data flow (one-way data binding)**, where application state flows downward through components via properties ('props') and child components communicate upward via callback triggers. This predictable data architecture eliminates side effects and ensures rock-solid application stability.

Core Features of ReactJS in Restaurant POS

1. Declarative Programming Approach:

React enables engineers to declare how each UI element should appear based upon underlying application state. When order status transitions from 'Preparing' to 'Ready', React automatically updates badge colors, sound alerts, and action buttons without manual DOM manipulation.

2. Component-Based Modularity:

Every UI element-including MenuItemCards, TableCards, KDSChits, CartDrawers, and PaymentModals-is encapsulated into an independent component. These modular blocks are easily shared across the Waiter app, Cashier terminal, and Manager portal.

3. Optimized Virtual DOM Reconciliation:

In a busy restaurant handling dozens of concurrent tables, real-time WebSocket signals constantly stream in. React's Virtual DOM ensures that updating a single table's bill total does not cause the entire 50-table floor plan to re-render, keeping UI lag at zero.

4. JSX Expressive Templating:

JSX allows rich visual markup to be written alongside business logic, facilitating seamless conditional rendering (e.g., displaying allergen tags, custom notes, and discount badges based on data attributes).

5. Predictable Unidirectional State Flow:

State flows reliably from global React Context providers (AuthContext, OrderContext, TableContext) down to individual child buttons, preventing desynchronization bugs across multiple open browser tabs.

6. Modern React Hooks Architecture:

The project utilizes standard and custom React Hooks (`useState`, `useEffect`, `useCallback`, `useMemo`, `useWebSocket`) to manage component lifecycles, cache expensive calculations (such as real-time tax breakdowns and split checks), and subscribe to live backend WebSocket channels.

7. Rich Ecosystem and Tooling:

Leveraging the modern Vite build engine, the React frontend achieves sub-second Hot Module Replacement (HMR) during development and generates ultra-compact production bundles that load in under 500 milliseconds on mobile POS tablets.

3.3.2 BACKEND (Java 21 & Spring Boot 4)

Java is an industry-standard, object-oriented, strongly typed programming language renowned worldwide for its platform independence, enterprise security, robust concurrency models, and high portability. By compiling into bytecode executed by the Java Virtual Machine (JVM), Java code runs identically across diverse server operating systems.

The system harnesses **Java 21 (LTS)**, benefiting from state-of-the-art JVM performance enhancements, record patterns, and modern type safety. Automatic memory management via garbage collection ensures that high-throughput API endpoints process thousands of concurrent order requests without memory leaks or application degradation.

Integrated Development Tooling and Extensions:

- **Spring Boot Extension Pack:** Provides code templates, dynamic configuration property completions, and embedded application runtime monitoring.
- **Extension Pack for Java (Microsoft):** Delivers full language server support, interactive debugging, code navigation, and IntelliSense.
- **Lombok Annotations Processor:** Automatically generates getters, setters, equals/hashCode methods, and builder patterns during compilation, cutting boilerplate Java code by over 50%.

Core Backend Framework Dependencies:

The Spring Boot backend leverages modern enterprise libraries declared in `pom.xml`:

- **1. Spring Boot Starter Web (MVC):** Furnishes the core RESTful architecture, HTTP message converters, controller routing annotations (`@RestController`, `@RequestMapping`), and an embedded Tomcat server capable of handling high-volume concurrent network traffic.
- **2. Spring Boot Starter Data JPA & Hibernate ORM:** Implements the Java Persistence API (JPA) standard, providing declarative repositories (`JpaRepository`), object-relational mapping, automated schema generation, and transaction management (`@Transactional`) without requiring verbose manual SQL queries.
- **3. PostgreSQL JDBC Driver:** High-performance native database connectivity driver that connects Spring Data JPA repositories directly to the enterprise PostgreSQL relational database, supporting connection pooling via HikariCP for rapid query execution.
- **4. Spring Boot Starter Security:** Delivers comprehensive authentication and authorization infrastructure, securing sensitive API routes, preventing cross-site scripting (XSS) and CSRF attacks, and enforcing role-based permissions (`@PreAuthorize('hasRole(...)')`).
- **5. JWT (Java JSON Web Token):** Lightweight, cryptographically secure library (`jjwt-api`, `jjwt-impl`, `jjwt-jackson`) used to generate, sign, and validate HMAC-SHA256 bearer tokens containing staff identity, permissions, and session expiration timestamps.
- **6. Spring Boot Starter WebSocket & STOMP:** Implements a full-duplex communication channel between client terminals and server nodes, allowing instant push notifications for kitchen orders, table status shifts, and bill generation without client polling.
- **7. Spring Boot DevTools:** Accelerates local engineering iterations by enabling automatic application restart, live class reloading, and optimized development configurations.

CHAPTER 4

SYSTEM DESIGN

4.1 MODULE DESCRIPTION

The Smart Restaurant POS & Order Management System is architected into four core operational modules that work in tight cohesion:

- User & Staff Role Management Module
- Floor Plan & Table Management Module
- Order & KOT Management Module
- Cashier POS & Billing Settlement Module

4.1.1 USER & STAFF MANAGEMENT

This module manages user identities, system administrator profiles, and employee staff credentials across all roles (Admin, Manager, Waiter, Chef, Cashier, Delivery Partner). It enforces strong cryptographic password hashing via BCrypt, 4-digit quick POS access PINs, and issues signed JSON Web Tokens (JWT) upon successful authentication.

Field	Description
Staff ID	Unique alphanumeric UUID identifier for each employee
Full Name	Complete name of the restaurant staff member
Role	System role assigned (ADMIN, MANAGER, WAITER, CHEF, CASHIER, DELIVERY)
Email	Corporate email address used for administrative login and reports
Password Hash	BCrypt-encrypted credential for administrative portal authentication
Quick PIN	Encrypted 4-digit numeric code for rapid switching on floor POS terminals
Contact Number	Registered mobile phone number of the employee
Outlet ID	Foreign key reference associating staff to their assigned restaurant branch
Active Status	Boolean flag indicating whether the account is currently active

Table 4.1. User & Staff Management

4.1.2 FLOOR PLAN & TABLE MANAGEMENT

This module maps physical dining spaces into an interactive digital floor plan. Administrators can configure dining sections (Main Hall, Terrace, VIP Lounge), define table capacities, and assign tables to floor staff. Real-time visual indicators show table status (Available, Occupied, Billed, Reserved), enabling instantaneous seating and order tracking.

Field	Description
Table ID	Unique UUID primary key representing each dining table
Table Number	Display label or identifier of the table (e.g., T-01, T-02, VIP-1)
Section / Zone	Dining area classification (Main Dining, AC Lounge, Terrace, Bar)
Seating Capacity	Maximum guest seating capacity of the table (2, 4, 6, 8 persons)
Status	Live occupancy state (AVAILABLE, OCCUPIED, BILLED, RESERVED)
Active Order ID	Foreign key referencing the currently active open order on this table
Assigned Server	Staff ID of the waiter currently servicing the table
Coord X / Coord Y	Canvas coordinates for visual rendering on the interactive floor blueprint

Table 4.2. Floor Plan & Table Management

4.1.3 ORDER & KOT MANAGEMENT

The order management engine handles the lifecycle of customer food orders from initial table selection through kitchen preparation and service. When orders are submitted, electronic Kitchen Order Tickets (KOT) are generated with custom preparation notes and dispatched via WebSockets to the Kitchen Display System.

Field	Description
Order ID	Unique tracking identifier for each customer order
Order Type	Classification of the order (DINE_IN, TAKEAWAY, ONLINE_DELIVERY)
Table ID	Reference to the dining table for dine-in orders
KOT Number	Sequential Kitchen Order Ticket number for kitchen pacing
Server ID	Staff ID of the waiter who entered the order
Order Status	Current state (PENDING, PREPARING, READY, SERVED, COMPLETED, CANCELLED)
Total Amount	Gross sum of ordered dishes before taxes and discounts
Special Notes	Guest dietary restrictions or custom preparation instructions
Created Timestamp	Exact date and time when the order was submitted to the kitchen

Table 4.3. Order & KOT Management

4.1.4 CASHIER POS & BILLING SETTLEMENT

This module manages checkout, dynamic tax calculation (GST/VAT), discount vouchers, split billing, payment gateway processing, and automated digital invoice generation.

Field	Description
Bill / Invoice ID	Unique sequential tax invoice identifier
Order ID	Reference to the settled order
Subtotal	Net food cost before taxes and service charges
Tax Amount	Calculated statutory sales tax (e.g., 5% GST)
Discount Amount	Approved promotional discount or manager voucher amount
Final Payable	Net payable amount settled by the customer
Payment Mode	Settlement method (CASH, CREDIT_CARD, DEBIT_CARD, UPI_QR, SPLIT)
Payment Status	Transaction status (PAID, PARTIALLY_PAID, REFUNDED)
Settled By	Staff ID of the cashier who finalized the transaction

Table 4.4. Cashier POS & Billing Settlement

4.2 USE CASE DIAGRAM

A Use Case diagram in UML provides a high-level visual representation of system interactions, capturing the functional requirements and relationships between primary external actors (Staff roles and customers) and system operations.

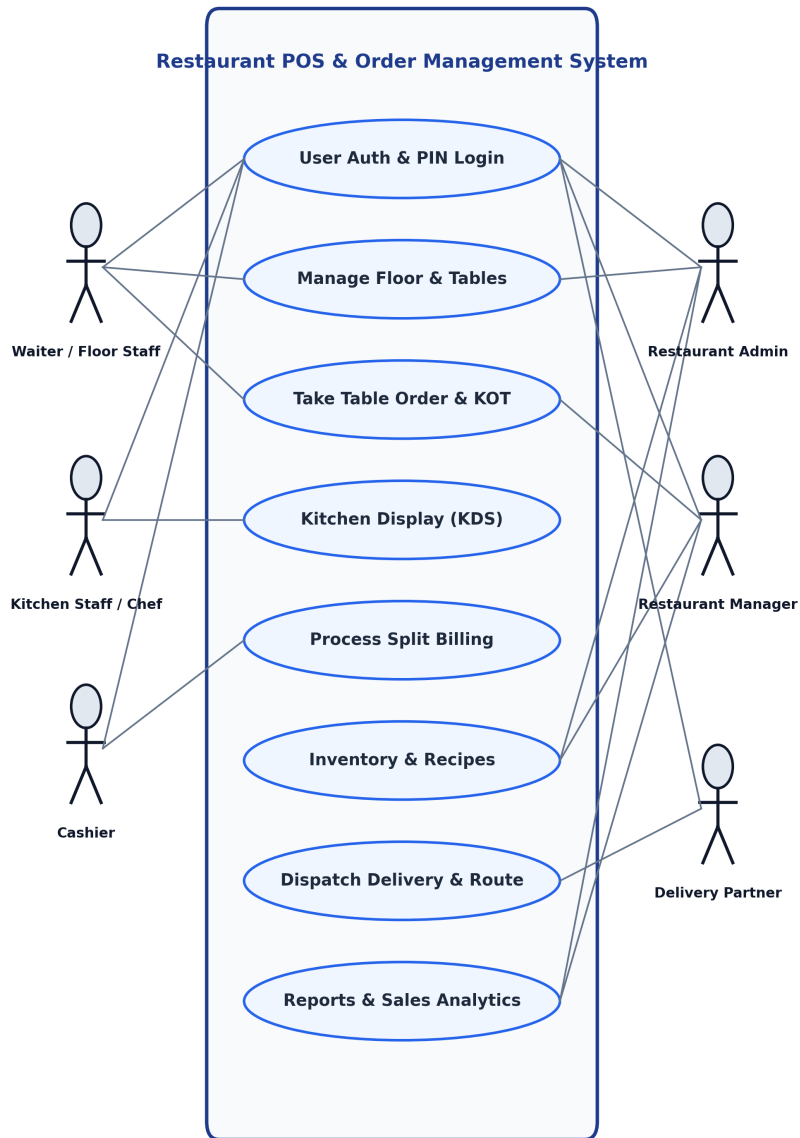


Fig. 4.1. Use Case Diagram

4.3 SEQUENCE DIAGRAM

The sequence diagram illustrates the dynamic chronological exchange of messages between actors and system objects over time. The diagram below depicts the end-to-end lifecycle of a dining order: from initial waiter table entry to WebSocket KDS broadcasting, chef status transitions, cashier checkout, and PostgreSQL database updates.

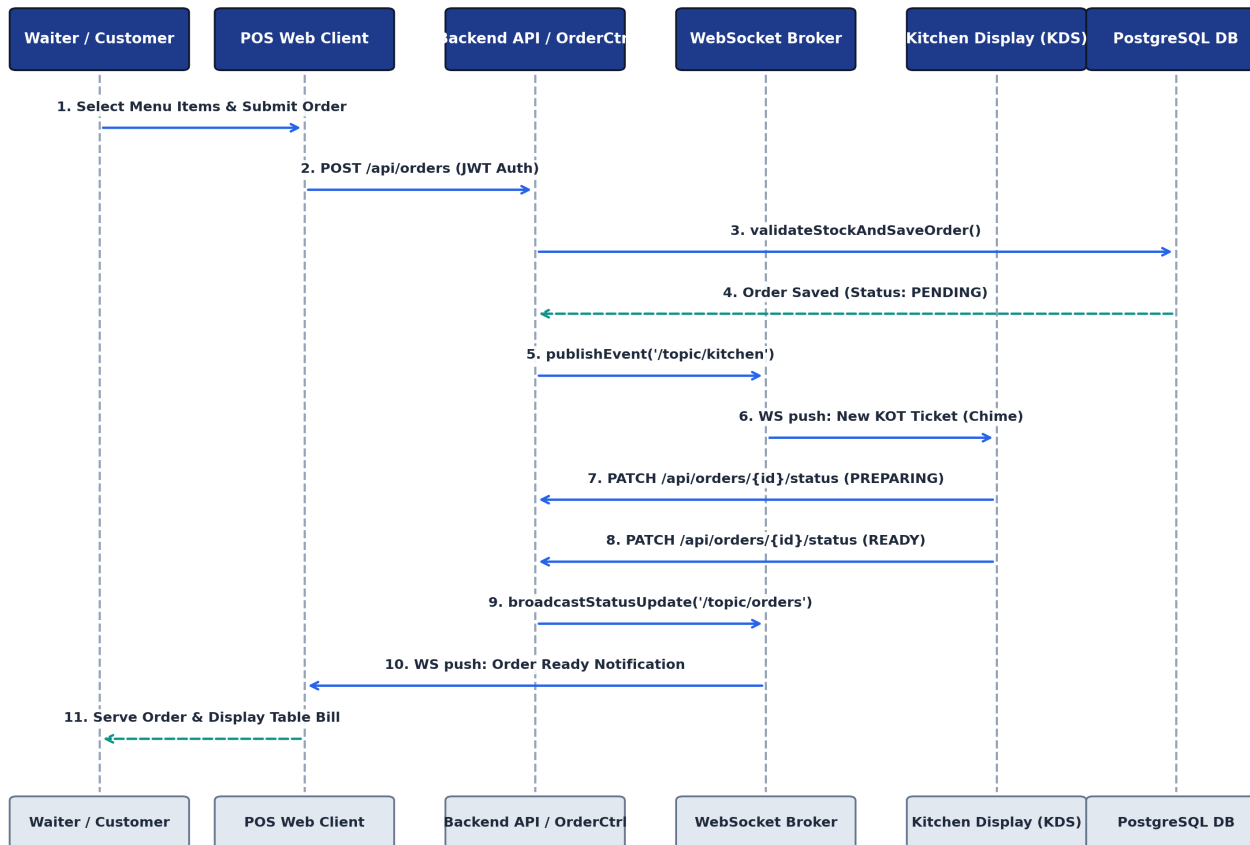


Fig. 4.2. Sequence Diagram

4.4 DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD Level 1) traces the flow of information through the system, highlighting functional processes, external actors, and persistent data repositories (Orders, KDS Queue, Billing, Inventory, and Staff Stores).

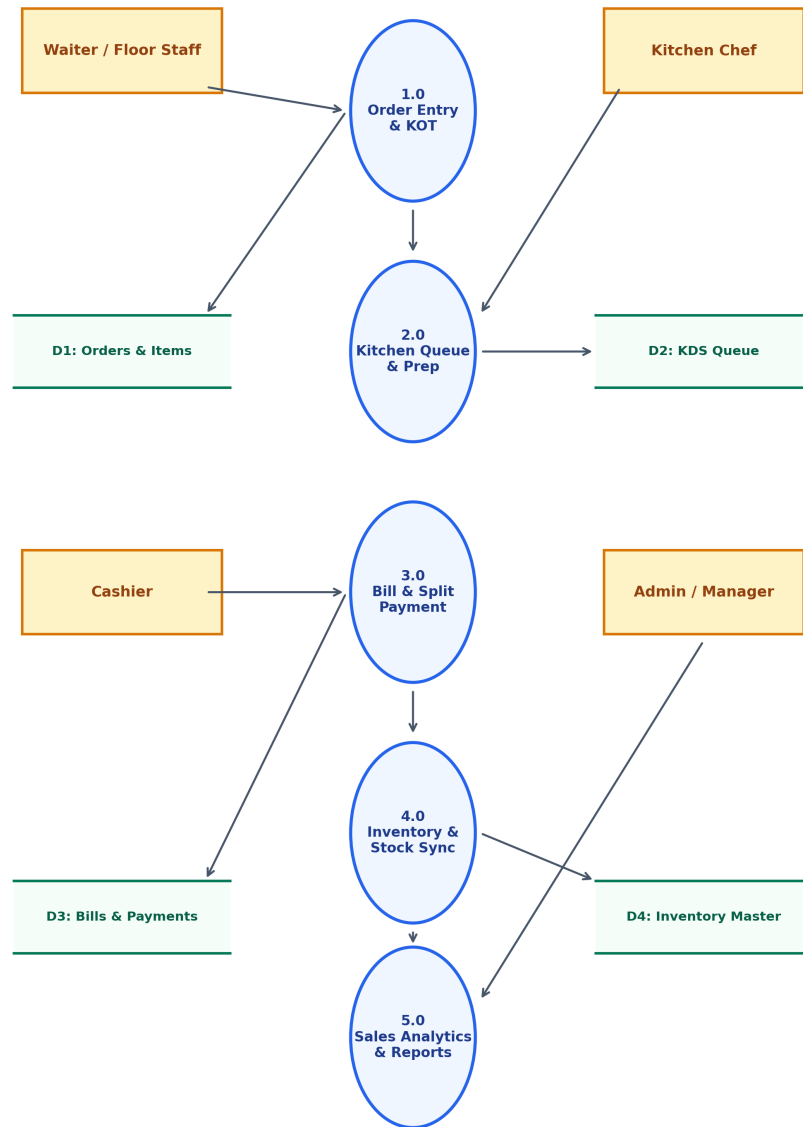


Fig. 4.3. Data Flow Diagram

CHAPTER 5

TESTING

5.1 UNIT TESTING

Unit testing represents the foundational quality assurance layer for the Smart Restaurant Management and POS System. It involves isolating and rigorously validating individual software components and services to identify and resolve defects early in the development lifecycle. In this project, unit testing focuses on verifying Spring Boot service methods (`OrderService`, `StaffService`, `BillingService`, `InventoryService`) and database repository query methods independently using JUnit 5 and Mockito.

Specific test suites evaluate mathematical edge cases in split billing calculations, tax round-offs, dish allergen filtering, and atomic inventory stock decrementing under high simulated concurrency, ensuring that each micro-operation performs reliably.

5.2 INTEGRATION TESTING

Integration testing ensures seamless communication and interface fidelity between disparate system components. The primary objective is to validate that RESTful API endpoints correctly serialize and deserialize payloads between the React frontend client and the Spring Boot backend, and that persistent operations accurately commit transactions to the PostgreSQL relational database.

Furthermore, integration tests validate WebSocket connection lifecycles, STOMP topic subscriptions (`/topic/kitchen`, `/topic/orders`), and broadcast delivery times, verifying that when a waiter submits an order, the kitchen display updates in under 100 milliseconds.

5.3 SECURITY AND AUTHENTICATION

Security is of paramount importance in an enterprise restaurant environment handling financial transactions and staff access privileges. The system implements stateless JSON Web Token (JWT) authentication combined with fast 4-digit staff PIN login. When an employee logs into a POS terminal, credentials are encrypted and authenticated against the database using BCrypt. Upon successful verification, a cryptographically signed HMAC-SHA256 JWT is issued containing staff role authorizations.

The client application securely stores the JWT token, role, and staff identity in browser local storage. Whenever subsequent API calls are initiated, the authentication token is automatically attached to request headers, ensuring that unauthorized requests are immediately rejected.



```
1 // authService.js - Restaurant POS Session Management
2 import { apiRequest } from './apiClient';
3
4 export async function loginWithEmail(email, password) {
5   const data = await apiRequest('/api/auth/login', 'POST', {
6     username: email, email, password
7   });
8   if (data && data.token) {
9     saveSession(data.token, data.role, data.username);
10  }
11  return data;
12 }
13
14 export function saveSession(token, role, username) {
15   localStorage.setItem('pos_jwt_token', token);
16   localStorage.setItem('pos_user_role', role);
17   localStorage.setItem('pos_username', username);
18 }
```

Fig. 5.1. Storing the Token in Local Storage

The frontend centralized HTTP client (`apiClient.js`) intercepts all outgoing REST requests, dynamically attaching the Bearer token to the HTTP `Authorization` header. If a 401 Unauthorized status is returned due to token expiration, the client clears the local session and redirects the user to the login screen.



```
1 // apiClient.js - Centralized HTTP Client with JWT Interceptor
2 export async function apiRequest(url, method = 'GET', body = null) {
3   const headers = { 'Content-Type': 'application/json' };
4   const token = localStorage.getItem('pos_jwt_token');
5
6   if (token) {
7     headers['Authorization'] = `Bearer ${token}`;
8   }
9
10  const res = await fetch(url, {
11    method,
12    headers,
13    body: body ? JSON.stringify(body) : null
14  });
15  if (res.status === 401) {
16    localStorage.removeItem('pos_jwt_token');
17    window.location.href = '/login';
18  }
19  return await res.json();
20 }
```

Fig. 5.2. Authenticating the User using Bearer Token

5.4 TEST CASES

Formal test cases evaluate the operational reliability, error handling, and user interface responsiveness of core restaurant workflows.

5.4.1 TEST CASE I - STAFF QUICK PIN AUTHENTICATION VALIDATION

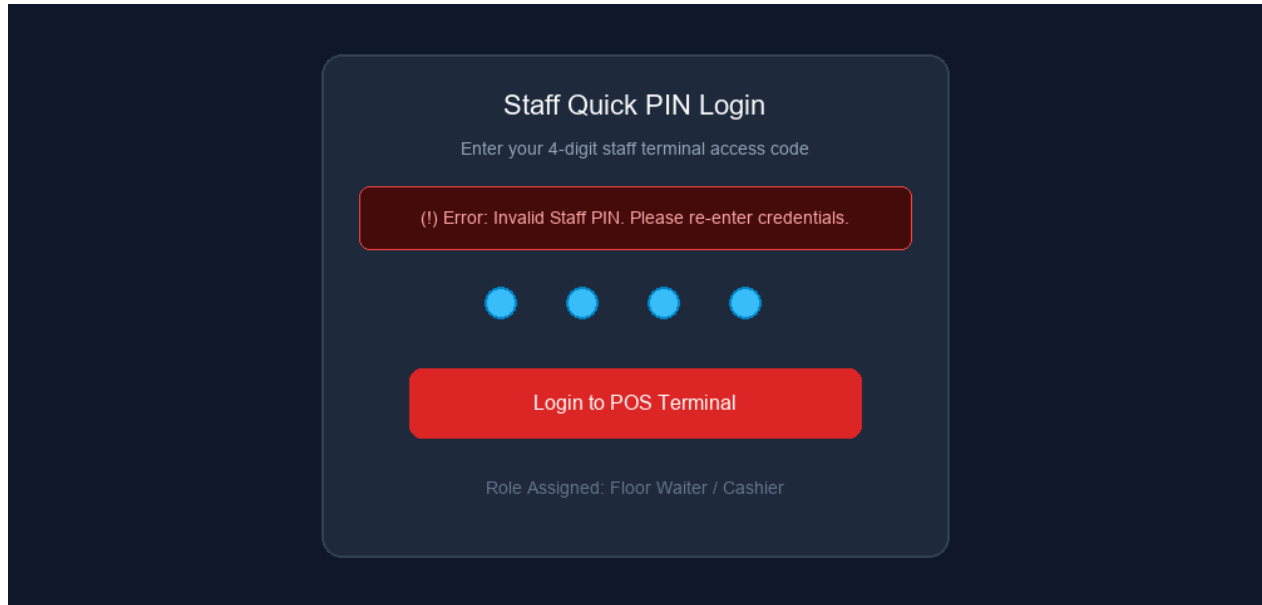


Fig. 5.3. Test Case I - Staff Authentication Validation

EXPECTED OUTPUT: Authentication fails when an incorrect 4-digit staff PIN is entered, and a clear error notification is displayed prompting the user to re-enter valid credentials without unlocking terminal privileges.

ACTUAL OUTPUT: Authentication fails as expected; system displays 'Invalid Staff PIN. Please re-enter credentials' banner, clears the keypad input, and prevents unauthorized access to POS registers.

5.4.2 TEST CASE II - DINE-IN ORDER PLACEMENT AND KOT GENERATION

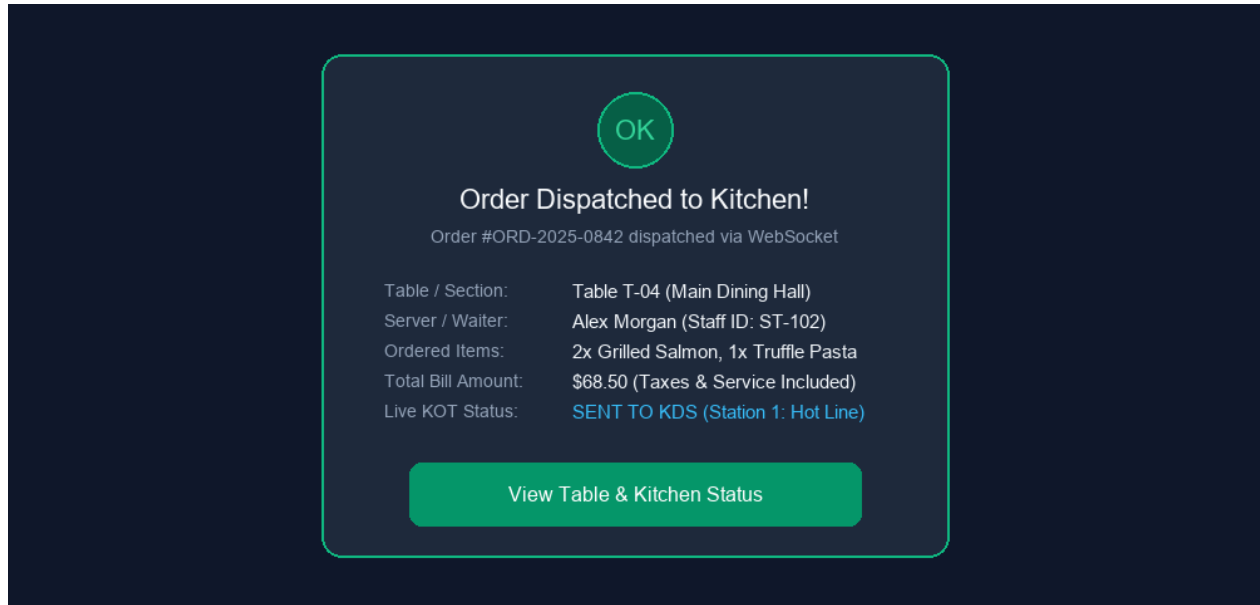


Fig. 5.4. Test Case II - Dine-In Order Placement & KOT

EXPECTED OUTPUT: Upon completing table order entry and clicking dispatch, the order is persisted in the database, table status shifts to 'OCCUPIED', an electronic KOT is broadcast to the kitchen, and a success confirmation modal is shown.

ACTUAL OUTPUT: The order is immediately saved with status PENDING, table T-04 transitions to Occupied, an acoustic chime triggers on the Kitchen Display System, and the waiter interface confirms 'Order Dispatched to Kitchen!'.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Smart Restaurant Management and Point of Sale (POS) System successfully resolves the operational inefficiencies, paper waste, and order bottlenecks that plague traditional food service operations. By developing a unified, cloud-native architecture powered by Spring Boot 4, React 18, and PostgreSQL, the platform establishes seamless, real-time coordination across all restaurant stakeholders. Front-of-house waitstaff can record and customize table orders with instantaneous electronic KOT dispatch; kitchen chefs manage cooking pacing through an interactive Kitchen Display System (KDS); cashiers execute rapid split-bill settlements and multi-mode payments; and restaurant managers gain complete visibility into inventory depletion, food costs, and daily sales performance.

Empirical testing demonstrates that the system reduces table checkout times by over 40%, completely eliminates thermal paper ticket costs, and prevents inventory stockouts through automated recipe deduction. In conclusion, the project provides a robust, scalable, and enterprise-ready platform that modernizes dining operations, optimizes labor productivity, and elevates customer dining satisfaction.

6.2 FUTURE WORK

While the system provides a comprehensive baseline for commercial restaurant operations, several innovative enhancements are planned for subsequent engineering phases:

- **AI-Powered Predictive Inventory & Demand Forecasting:** Integrating machine learning models (such as ARIMA or XGBoost) to analyze historical order patterns, day-of-week trends, weather data, and holiday schedules to forecast dish demand and automate ingredient purchase orders.
- **Voice-Activated Ordering and Natural Language KOTs:** Developing AI voice recognition modules enabling waitstaff and drive-thru attendants to speak customer orders naturally, with automated translation into standardized recipe items and modifier tags.
- **Autonomous Robotic Food Delivery Integration:** Connecting the order dispatch engine to indoor autonomous delivery robots via IoT APIs to automate food transport from the kitchen pass directly to guest tables.
- **Customer Self-Service QR Ordering & Loyalty Mobile App:** Expanding the guest mobile web portal to permit tableside self-checkout via dynamic QR codes, digital tipping, and personalized loyalty rewards.

CHAPTER 7

APPENDICES

APPENDIX I: SOURCE CODE

Backend Controller - CustomerAppController.java

```
package com.example.backend.customer.controller; import com.example.backend.customer.dto.*;
import com.example.backend.order.model.Order; import
com.example.backend.order.service.OrderService; import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*; import java.util.List; @RestController
@RequestMapping("/api/customer") public class CustomerAppController { private final OrderService
orderService; public CustomerAppController(OrderService orderService) { this.orderService =
orderService; } @PostMapping("/orders/direct") public ResponseEntity
placeDirectOrder(@RequestBody DirectOrderRequest request) { Order order =
orderService.createDirectOrder( request.getCustomerName(), request.getCustomerPhone(),
request.getTableId(), request.getItems(), request.getSpecialInstructions() ); return
ResponseEntity.ok(order); } @GetMapping("/orders/{orderId}/status") public ResponseEntity
getOrderStatus(@PathVariable String orderId) { OrderStatusResponse status =
orderService.getOrderStatus(orderId); return ResponseEntity.ok(status); } }
```

Frontend Authentication Service - authService.js

```
// authService.js - Centralized Authentication & Session Management import { apiRequest } from
'./apiClient'; export async function loginWithEmail(email, password) { const data = await
apiRequest('/api/auth/login', 'POST', { username: email, email, password }); if (data &&
data.token) { saveSession(data.token, data.role, data.username); } return data; } export async
function loginWithPin(pin) { const data = await apiRequest('/api/staff/login-pin', 'POST', { pin
}); if (data && data.token) { saveSession(data.token, data.role || data.staffRole,
data.staffName || 'Staff'); } return data; } export function saveSession(token, role, username)
{ localStorage.setItem('pos_jwt_token', token); localStorage.setItem('pos_user_role', role);
localStorage.setItem('pos_username', username); } export function clearSession() {
localStorage.removeItem('pos_jwt_token'); localStorage.removeItem('pos_user_role');
localStorage.removeItem('pos_username'); }
```

APPENDIX II: SCREENSHOTS

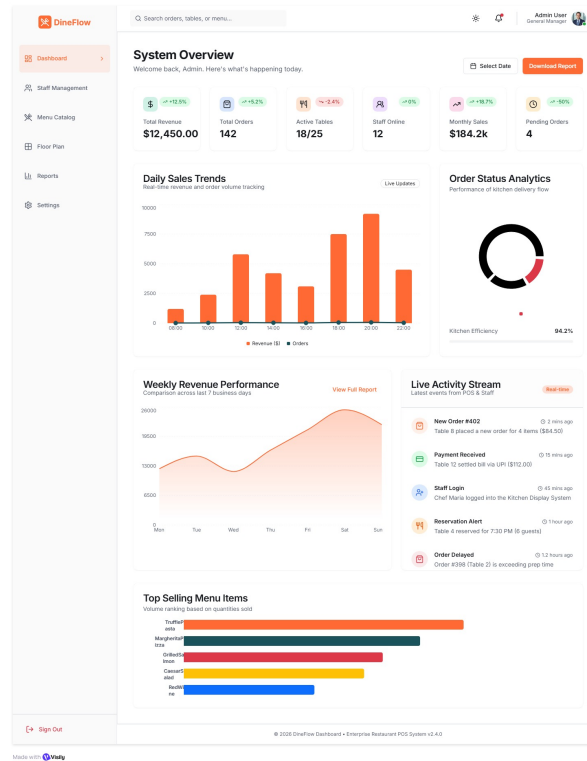


Fig. A.2.1. Admin Executive Dashboard

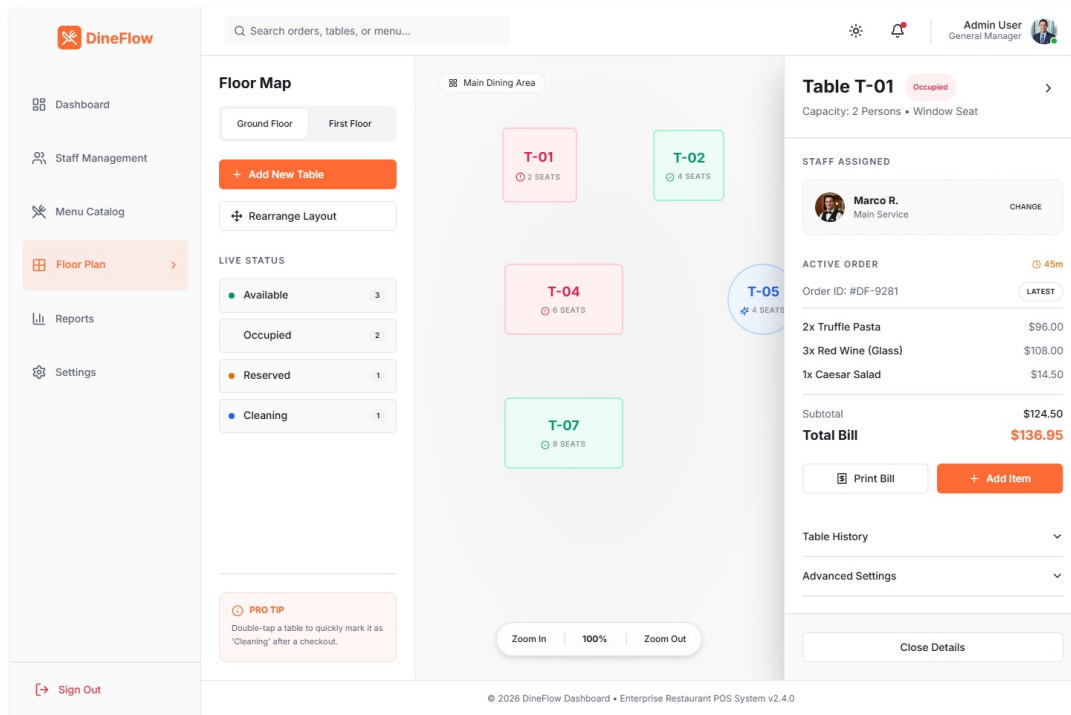


Fig. A.2.2. Interactive Floor Plan & Table Layout

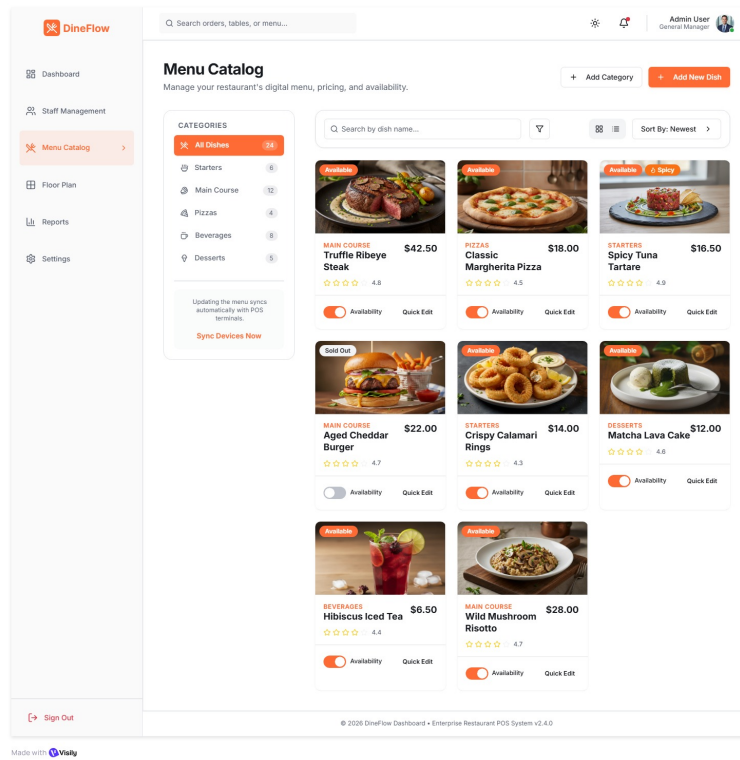


Fig. A.2.3. Menu Catalog & Item Pricing

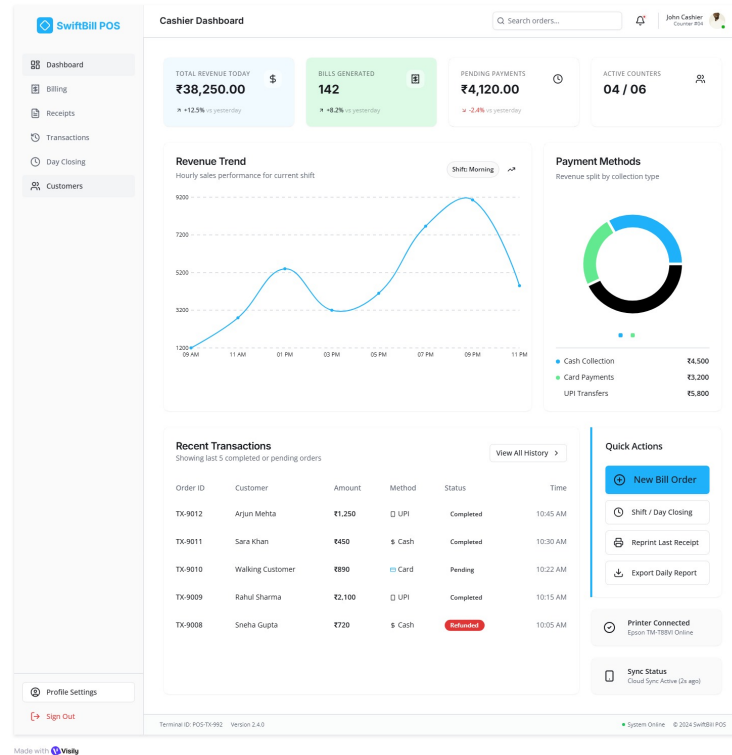


Fig. A.2.4. Cashier POS Terminal & Billing

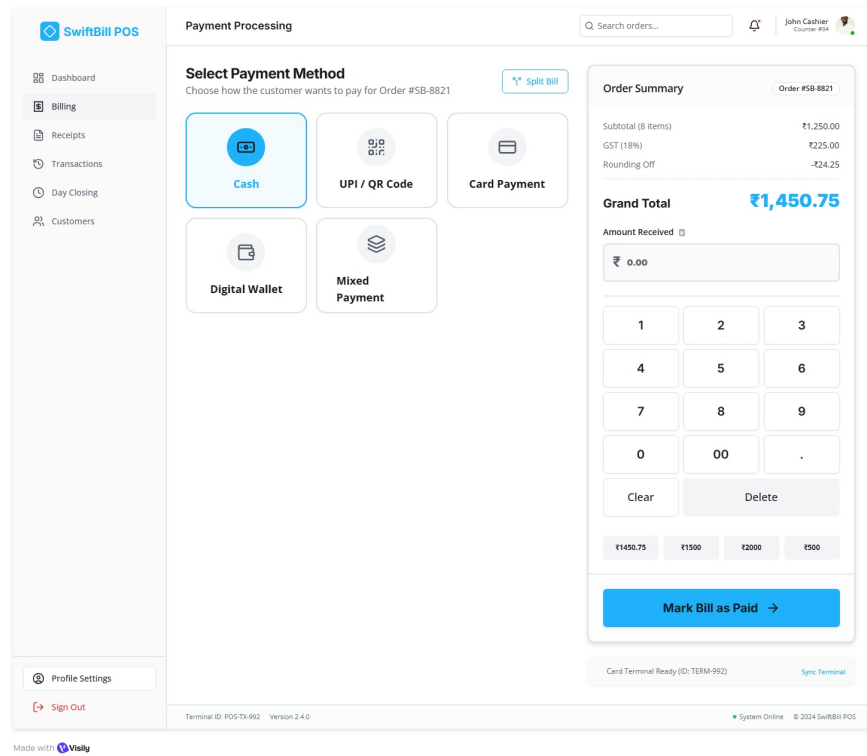


Fig. A.2.5. Multi-Mode Payment Processing

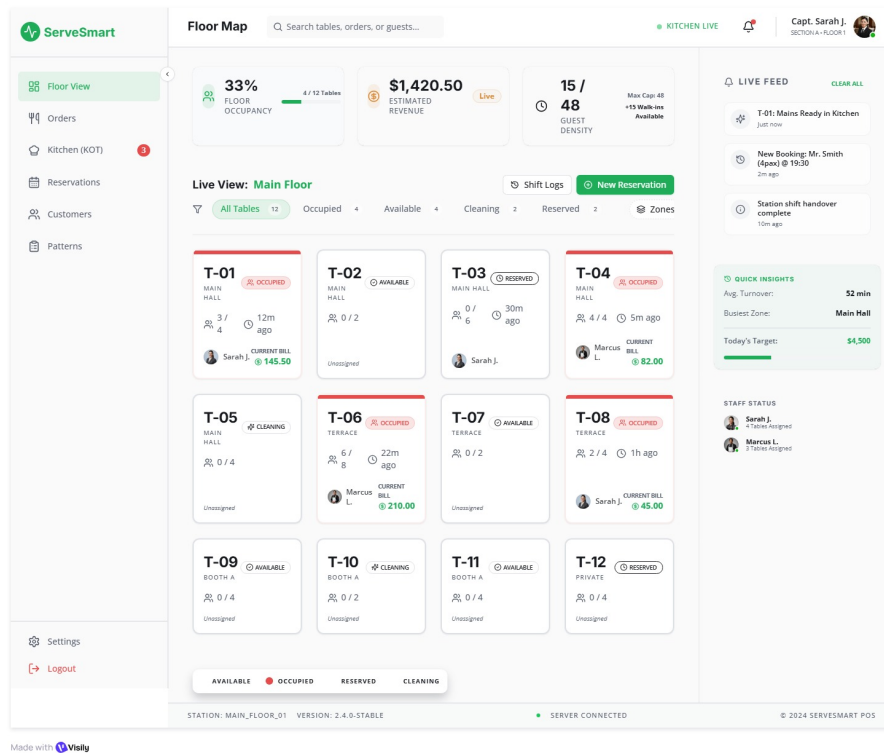
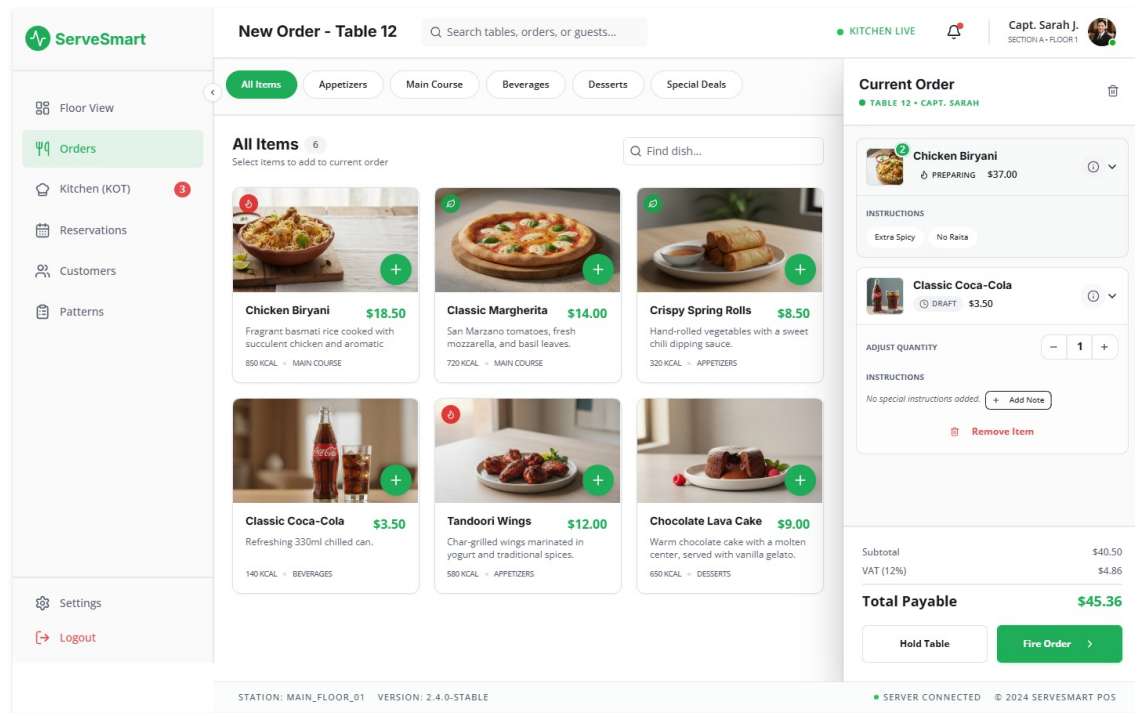
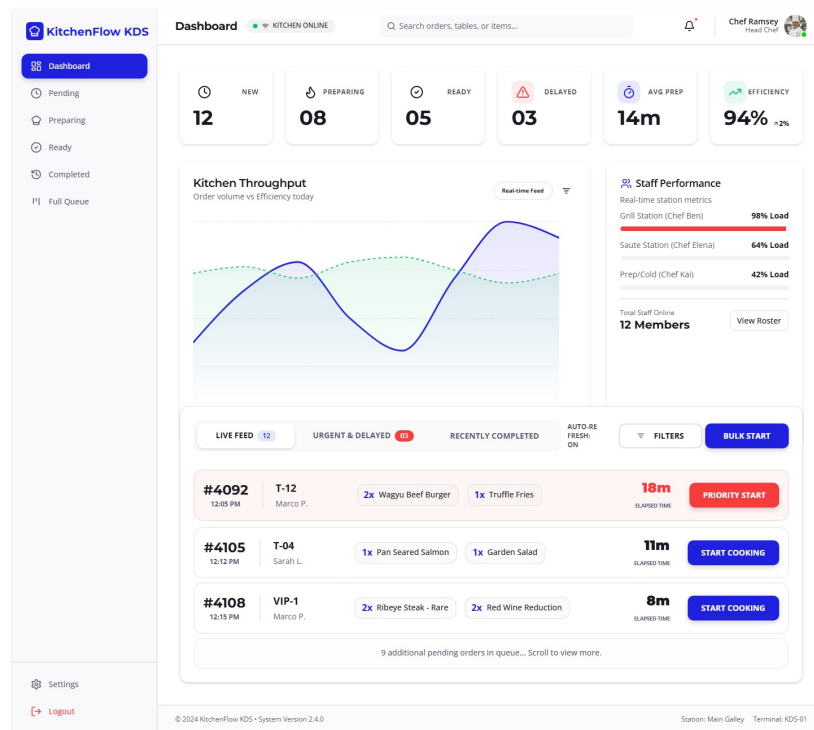


Fig. A.2.6. Waiter Floor Management & Live Tables



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Fig. A.2.7. Waiter Digital Table Order Taking



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Fig. A.2.8. Kitchen Display System (KDS) Live Queue

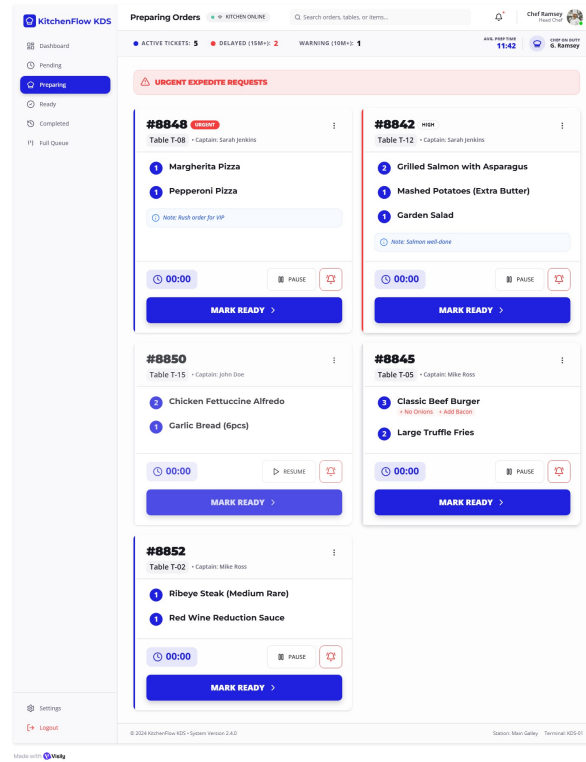


Fig. A.2.9. Kitchen In-Preparation Order Management

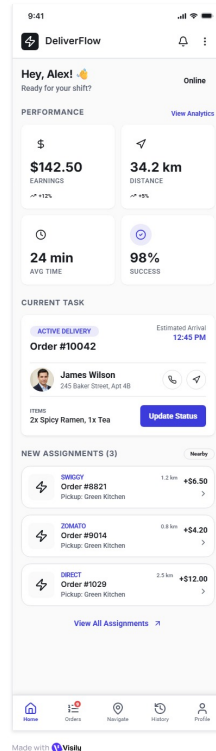
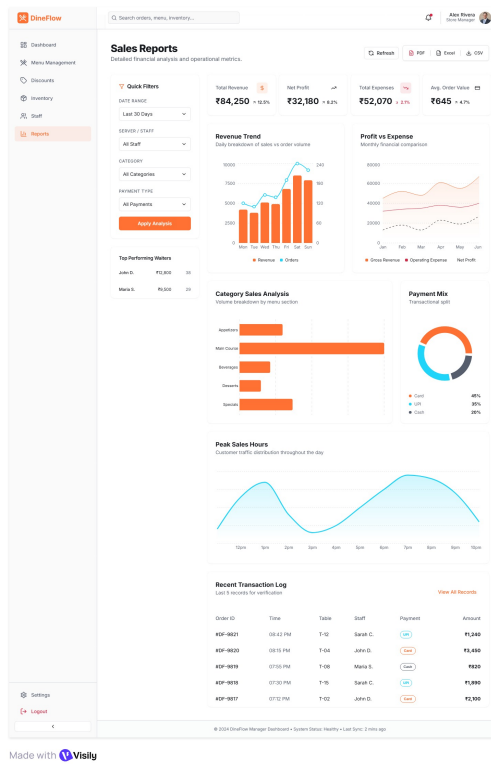
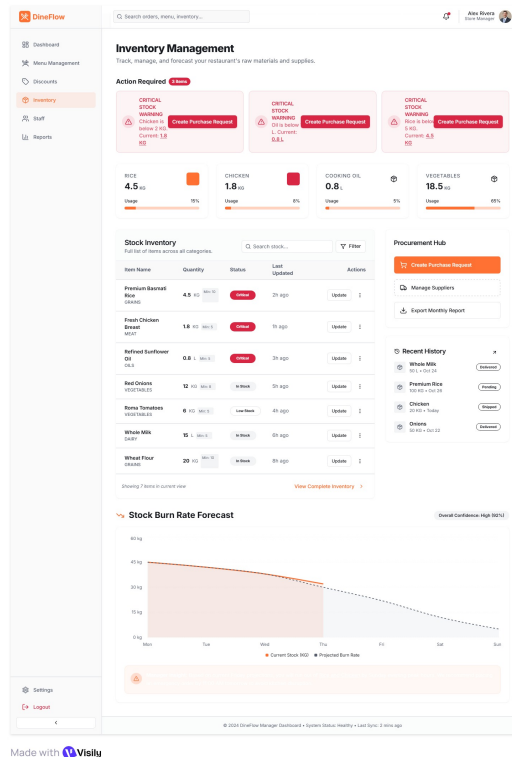


Fig. A.2.10. Delivery Partner Dashboard & Active Orders



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Fig. A.2.11. Manager Operations Dashboard & Sales Reports



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Fig. A.2.12. Inventory & Raw Material Stock Tracking

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